

Aliah University

End-Semester Supplementary Examination (Autumn Semester) - 2021

(For 3rd Year 5th Semester B.Tech (CSE))

Paper Name: Operating Systems

Full Marks: 80

Paper Code: CSE301

Time: 3 hrs

Group-A

(Answer all questions)

10 × 1 = 10

1. What is a Process?
2. What does PCB contain?
3. What do you mean by inter-process communication?
4. What is the maximum length of the filename in DOS?
5. What else is a command interpreter called?
6. What is deadlock?
7. Define Seek Time and Latency Time.
8. Define waiting time.
9. When does page fault occur?
10. What is the fence register used for?

Group-B

(Answer any six questions)

6 × 5 = 30

1. A) Explain lifecycle of a process with a suitable diagram. B) What is PCB and explain it. (3+2= 5)
2. Consider the following table and find out i) average waiting time ii) turnaround time for preemptive-Priority CPU scheduling algorithms. (Higher number represents higher priority) (5)

Process	Arrival Time	Burst Time	Priority
P1	0	4	2
P2	1	3	3
P3	2	1	4
P4	3	5	5
P5	4	2	5
3. A) Differentiate between a Process and a Program. B) Differentiate distributed systems from multiprocessor system? (2+3 = 5)
4. A) What is scheduler? B) What are different types of schedulers? Explain it. (1+4 = 5)
5. A) Define independent process and dependent process. B) What is IPC? Name different methods of it. (2+3 = 5)
6. Explain about resource allocation graph (RAG)? (5)
7. A) What are the advantages of Contiguous Allocation? B) How protection is done during paging? (2+3 =5)

Group-C

(Answer any four questions)

4 × 10 = 40

1. A) Explain race condition with a suitable example. B) What is critical-section problem? 8+2 = 10
2. A) Differentiate SCAN and C-SCAN scheduling algorithm. 2+ 8 = 10
B) A disk queue has the following requests to read tracks –
87, 170, 40, 150, 36, 72, 66, 15

If the disk head is initially at cylinder 60, then what is the total number of head movements needed to satisfy these requests for the i) FCFS and ii) SSTF disk- scheduling algorithms?

3. A) Consider the following reference string: **8+2=10**
 0, 1, 2, 3, 0, 1, 2, 3, 0, 1, 2, 3, 4, 5, 6, 7
 How many page faults will occur for (i) FIFO and (ii) LRU page replacement algorithms?
 Assuming **three memory** frames are there and all are initially empty.
 B) What is Belady's Anomaly?

4. A) What is paging? Explain the paging methods with a suitable diagram. **(6+2+2 = 10)**
 B) What are advantages of TLB with paging over paging technique?
 C) What is demand paging?

5. A) Define and explain necessary conditions of deadlock. B) Explain resource-allocation graph with request edge and assignment edge.

C) Consider the following snapshot of a system: **3+3+4 = 10**

Process	Allocation				Max				Available			
	A	B	C	D	A	B	C	D	A	B	C	D
P0	0	0	1	2	0	0	1	2	1	5	2	0
P1	1	0	0	0	1	7	5	0				
P2	1	3	5	4	2	3	5	6				
P3	0	6	3	2	0	6	5	2				
P4	0	0	1	4	0	6	5	6				

- i) What is the content of the matrix Need?
 ii) If a request from process P1 arrives for (0,4,2,0), can the request be granted immediately?